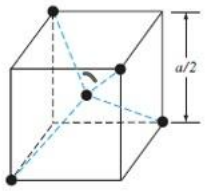
Homework 1 Due on 3/11/2026

Note:

- You may need to use Matlab or other software for this homework.
- You are not allowed to copy from others' solution or any solution on line.

1. Please create PowerPoint slides and a 10-minute video introducing the Ultraviolet Catastrophe in blackbody radiation (Optional, 50 pts, the ppt and video can be sent to TA directly)
2. Ge has the same crystal structure (diamond) as Si, with a lattice constant of a = 5.64 Angstroms = 0.564 nm. Find the atomic density (atoms/cm^3) and the spacing between nearest-neighbor atoms in Ge. Recall that 1 nm = 1x10^{-7} cm. (10pts)
3. Please calculate the angle between the tetrahedral bonds of diamond below. (10pts)

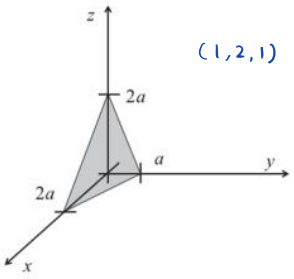


4. Gallium arsenide (GaAs) has a zinc-blende crystal structure. Answer the following questions about GaAs. (Assume a lattice spacing of a = 5.65 Angstroms = 0.565 nm.) (20pts)
   a) What is the density of GaAs in g/cm^3?
   b) How many atoms/cm^3 are there in GaAs?
   c) How many valence electrons per cm^3 are there in GaAs?
   d) What is the closest spacing between adjacent arsenic atoms in GaAs?
5. Silicon (Si) has a diamond crystal structure. Answer the following questions about Si. (Assume a lattice spacing of a = 5.42 Angstroms.) (20pts)
   a) Compute the density of Si atoms per cm^2 on {100} planes.
   b) Treat atoms as rigid spheres with radii equal to one-half of the distance between nearest neighbors. Compute the percentage of volume occupied by the Si atoms.
6. Sketch the following planes or directions. (20pts)
   a) (011)
   b) [110]
   c) (302)
   d) $(\bar{1}2\bar{1})$
7. Determine the Miller indices for the following planes along with the directions normal to each plane. (20pts)

a)

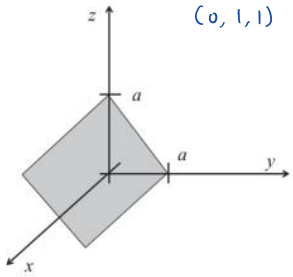
$\frac{1}{2}$ 1 $\frac{1}{2}$

(1, 2, 1)



b)

(0, 1, 1)

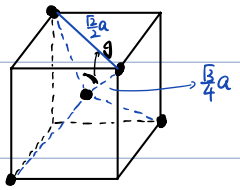


2. $a = 5.64\,\text{Å} = 5.64 \times 10^{-8}\,\text{cm}$

atomic density: $\rho = \frac{8}{a^3} = 4.46 \times 10^{22}\ \text{atoms/cm}^3$

spacing between neavest-neighbor atoms: $d = \frac{\sqrt{3}}{4}a \approx 2.44 \times 10^{-8}\,\text{cm} = 0.244\,\text{nm}$

3. Let θ be the angle between the tetrahedral bonds.



$$\cos\theta = \frac{2\cdot(\frac{\sqrt{3}}{4}a)^2 - (\frac{\sqrt{2}}{2}a)^2}{2\cdot(\frac{\sqrt{3}}{4}a)^2} = -\frac{1}{3}$$

$\theta = \arccos(-\frac{1}{3})$

Since $\theta \in [0, 360°]$. $\theta \approx 109.47°$

4. a) Atom mass: $A_{Ga} = 69.723\ \text{g/mol}$. $A_{As} = 74.922\ \text{g/mol}$.

density of GaAs: $\rho = \frac{4}{N_A}\frac{(A_{Ga}+A_{As})}{a^3} \approx 5.33\ \text{g/cm}^3$

b) autom density: $\rho = \frac{8}{a^3} \approx 4.43 \times 10^{22}\ \text{atoms/cm}^3$

c) valence electrons per lattice $= 4\times3 + 4\times5 = 32$

valence electrons density: $\rho = \frac{32}{a^3} \approx 1.78 \times 10^{23}$ valence electrons/cm^3

d) the closet spacing between adjacent arsenic atoms: $d = \frac{\sqrt{2}}{2}a \approx 3.99 \times 10^{-8}\,\text{cm}$

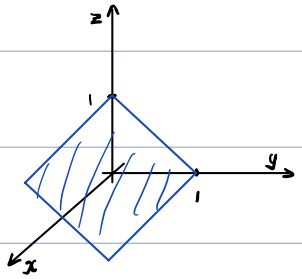
5. a) density of Si atoms on {100} planes : $\rho = \frac{2}{a^2} \approx 6.81 \times 10^{14}$ atoms/cm^2

b) distance between nearest neighbors : $d = \frac{\sqrt{3}}{4}a \approx 2.35 \times 10^{-8}$ cm

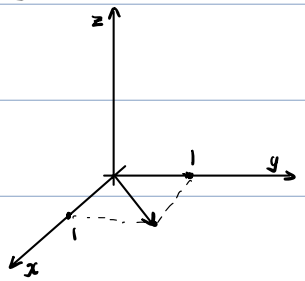
$r = \frac{d}{2} \approx 1.17 \times 10^{-8}$ cm

$\text{percentage} = \frac{8 \times \frac{4}{3}\pi r^3}{a^3} \times 100\% \approx 33.71\%$

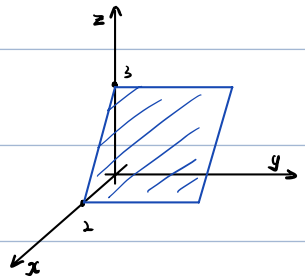
6. a) (011)



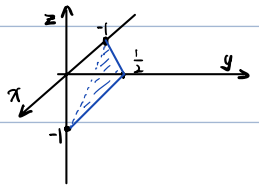
b) [110]



c) (302)



d) $(\bar{1}2\bar{1})$



7. a) Miller indices : (121)

directions : [121]

b) Miller indices : (011)

directions : [011]